

The recognition that inflammation plays a central role in atherosclerosis and thrombus formation is also influencing the treatment of cardiovascular disease. For example, aspirin, which inhibits the inflammatory response, has been found to help prevent the recurrence of heart attacks and stroke.

Hypertension (high blood pressure) is yet another contributor to heart attack and stroke. According to one hypothesis, chronic high blood pressure damages the endothelium that lines the arteries, promoting plaque formation. The usual definition of hypertension in adults is a systolic pressure above 140 mm Hg or a diastolic pressure above 90 mm Hg. Fortunately, hypertension can often be prevented and controlled by quitting any use of tobacco products, changing the diet, increasing exercise, taking medication, or a combination of these approaches.

CONCEPT CHECK 42.4

1. Explain why a physician might order a white cell count for a patient with symptoms of an infection.
2. Clots in arteries can cause heart attacks and strokes. Why, then, does it make sense to treat people with hemophilia by introducing clotting factors into their blood?
3. **WHAT IF?** Nitroglycerin (the key ingredient in dynamite) is sometimes prescribed for heart disease patients. Within the body, nitroglycerin is converted to nitric oxide (see Concept 42.3). Why would you expect nitroglycerin to relieve chest pain caused by narrowing of the cardiac arteries?
4. **MAKE CONNECTIONS** How do stem cells from the bone marrow of an adult differ from embryonic stem cells (see Concept 20.3)?

For suggested answers, see Appendix A.

CONCEPT 42.5

Gas exchange occurs across specialized respiratory surfaces

In the remainder of this chapter, we will focus on the process of **gas exchange**. Although this process is often called respiratory exchange or respiration, it should not be confused with the energy transformations of cellular respiration. Gas exchange is the uptake of molecular O_2 from the environment and the discharge of CO_2 to the environment.

Partial Pressure Gradients in Gas Exchange

To understand the driving forces for gas exchange, we must consider **partial pressure**, which is simply the pressure exerted by a particular gas in a mixture of gases. Determining partial pressures enables us to predict the net movement of a gas at an exchange surface: A gas always undergoes net diffusion from a region of higher partial pressure to a region of lower partial pressure.

To calculate partial pressures, we need to know the pressure that a gas mixture exerts and the fraction of the mixture represented by a particular gas. Let's consider O_2 as an example. At sea level, the atmosphere exerts a downward force equal to that of a column of mercury (Hg) 760 mm high. Atmospheric pressure at sea level is thus 760 mm Hg. Since the atmosphere is 21% O_2 by volume, the partial pressure of O_2 is 0.21×760 , or about 160 mm Hg. This value is called the **partial pressure** of O_2 (abbreviated P_{O_2}) because it is the part of atmospheric pressure contributed by O_2 . The partial pressure of CO_2 (abbreviated P_{CO_2}) is much, much less, only 0.29 mm Hg at sea level.

Partial pressures also apply to gases dissolved in a liquid, such as water. When water is exposed to air, an equilibrium state is reached such that the partial pressure of each gas in the water equals the partial pressure of that gas in the air. Thus, water exposed to air at sea level has a P_{O_2} of 160 mm Hg, the same as in the atmosphere. However, the **concentrations** of O_2 in the air and water differ substantially because O_2 is much less soluble in water than in air (**Table 42.1**). Furthermore, the warmer and saltier the water is, the less dissolved O_2 it can hold.

Table 42.1 Comparing Air and Water as Respiratory Media

	Air (Sea Level)	Water (20°C)	Air-to- Water Ratio
O_2 partial pressure	160 mm	160 mm	1:1
O_2 concentration	210 ml/L	7 ml/L	30:1
Density	0.0013 kg/L	1 kg/L	1:770
Viscosity	0.02 cP	1 cP	1:50

Respiratory Media

The conditions for gas exchange vary considerably, depending on whether the respiratory medium—the source of O_2 —is air or water. As already noted, O_2 is plentiful in air, making up about 21% of Earth's atmosphere by volume. As shown in Table 42.1, air is much less dense and less viscous than water, so it is easier to move and to force through small passageways. As a result, breathing air is relatively easy and the exchange does not need to not be particularly efficient. Humans, for example, extract only about 25% of the O_2 in inhaled air.

Water is a much more demanding gas exchange medium than air. The amount of O_2 dissolved in a given volume of water varies but is always less than in an equivalent volume of air. Water in many marine and freshwater habitats contains only about 7 mL of dissolved O_2 per liter, a concentration roughly 30 times less than in air. Water's lower O_2 content, greater density, and greater viscosity mean that aquatic animals such as fishes and lobsters must expend considerable energy to carry out gas exchange. In the context of